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EXPERIMENT 3
Morphological Analysis using NLTK

Aim

To perform basic morphological analysis of words in Python using the NLTK library by tokenizing text, assigning part-of-speech tags, and finding the base forms of words.

Algorithm

• Import the required NLTK functions and the WordNet lemmatizer.

• Download the required NLTK resources if they are not already installed.

• Read a sentence from the user.

• Tokenize the sentence into individual words.

• Assign part-of-speech tags to the tokens.

• Convert the words to WordNet-compatible part-of-speech categories.

• Lemmatize each word and display the word, its tag, and its base form.

Code

```
import nltk
from nltk import word_tokenize, pos_tag
from nltk.stem import WordNetLemmatizer

nltk.download("punkt")
nltk.download("averaged_perceptron_tagger")
nltk.download("wordnet")
nltk.download("omw-1.4")

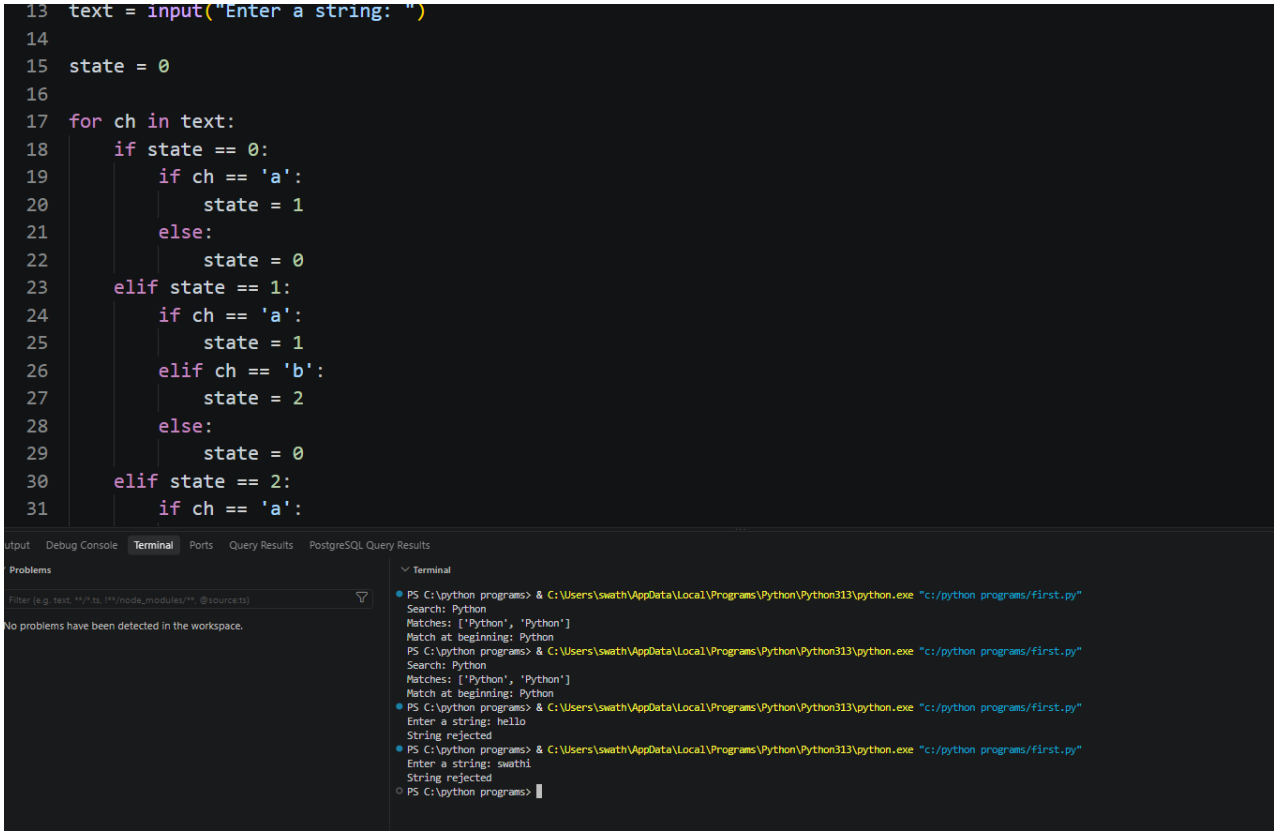
text = input("Enter a sentence: ")
words = word_tokenize(text)
tags = pos_tag(words)
lemmatizer = WordNetLemmatizer()

for word, tag in tags:
    if tag.startswith("V"):
        pos = "v"
    elif tag.startswith("J"):
        pos = "a"
    elif tag.startswith("R"):
        pos = "r"
    else:
        pos = "n"

    base = lemmatizer.lemmatize(word, pos)
    print(word, "->", base, "(", tag, ")")
```

Output

Output screenshot provided:



Result

The program demonstrates the basic steps of morphological analysis using NLTK, including tokenization, part-of-speech tagging, and lemmatization to identify base forms of words.